PROPOSITION QC COMPARE IN STACKS-PERFECT.TEX

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter stacks-perfect.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate is from the corrected band-I follow-on screen (line C) of the extractor's stacks-candidates line: its author label is absent from the live master in every form searched and it belongs to none of the earlier judged pools. The H2 judgement and its reason were made in this module by reading the author's own statement and complete proof in the fixed checkout. Original candidate key: stacks:stacks-perfect.tex:proposition-QC-compare.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 0H0X: sources/stacks-sheaves.tex, source lines 4964–4964. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 0H0C: sources/stacks-sheaves.tex, source lines 4510–4510. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07BA: sources/stacks-perfect.tex, source lines 862–862. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original remark 0H11: sources/stacks-perfect.tex, source lines 909–909. Navigation pointer used in the counted span; the span contains the pointer

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Source reference. Original lemma 0GZS: sources/sites-cohomology.tex, source lines 11608–11608. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0H0E: sources/stacks-sheaves.tex, source lines 4534–4534. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 08JA: sources/sites-cohomology.tex, source lines 8874–8874. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0H0T: sources/sites-cohomology.tex, source lines 11678–11678. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0H13: sources/stacks-perfect.tex, source lines 1154–1154. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0H14: sources/stacks-perfect.tex, source lines 1176–1176. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0H15: sources/stacks-perfect.tex, source lines 1197–1197. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let $\mathcal{X}$ be an algebraic stack. Then $QC(\mathcal{X})$ is canonically equivalent to $D_{QCoh}(\mathcal{O}_\mathcal{X})$.*

Proof. By Sheaves on Stacks, Lemma 0H0X pullback by the comparison morphism $\epsilon : \mathcal{X}_{affine,fppf} \to \mathcal{X}_{affine}$ identifies $QC(\mathcal{X})$ with a full subcategory $Q_\mathcal{X} \subset D(\mathcal{X}_{affine,fppf}, \mathcal{O})$. Using the equivalence of ringed topoi in Sheaves on Stacks, Equation (0H0C) we may and do view $Q_\mathcal{X}$ as a full subcategory of $D(\mathcal{X}_{fppf}, \mathcal{O})$.

Similarly by Lemma 07BA and Remark 0H11 we find that $D_{QCoh}(\mathcal{O}_\mathcal{X})$ may be viewed as the left orthogonal $\mathcal{A}$ of the left admissible subcategory $D_{Parasitic \cap LQCoh^{fbc}}(\mathcal{O}_\mathcal{X})$ of $D_{LQCoh^{fbc}}(\mathcal{O}_\mathcal{X})$.

To finish we will show that $Q_\mathcal{X}$ is equal to $\mathcal{A}$ as subcategories of $D(\mathcal{X}_{fppf}, \mathcal{O})$.

Step 1: $Q_{\mathcal{X}}$ is contained in $D_{LQCoh^{fbc}}(\mathcal{O}_{\mathcal{X}})$. An object K of $Q_{\mathcal{X}}$ is characterized by the property that K, viewed as an object of $D(\mathcal{X}_{affine,fppf}, \mathcal{O})$ satisfies $R\epsilon_* K$ is an object of $QC(\mathcal{X}_{affine}, \mathcal{O})$. This in turn means exactly that for all morphisms $x \to x'$ of $\mathcal{X}_{affine}$ the map

$$R\Gamma(x', K) \otimes^{\mathbf{L}}_{\mathcal{O}(x')} \mathcal{O}(x) \longrightarrow R\Gamma(x, K)$$

is an isomorphism, see footnote in statement of Cohomology on Sites, Lemma 0GZS. Now, if $x' \to x$ lies over a flat morphism of affine schemes, then this means that

$$H^i(x', K) \otimes_{\mathcal{O}(x')} \mathcal{O}(x) \cong H^i(x, K)$$

This clearly means that $H^i(K)$ is a sheaf for the étale topology (Sheaves on Stacks, Lemma 0H0E) and that it has the flat base change property (small detail omitted).

Step 2: $Q_{\mathcal{X}}$ is contained in $\mathcal{A}$. To see this it suffices to show that for K in $Q_{\mathcal{X}}$ we have $\mathrm{Hom}(K, P) = 0$ for all P in $D_{Parasitic \cap LQCoh^{fbc}}(\mathcal{O}_{\mathcal{X}})$. Consider the object

$$H = R\mathcal{H}om_{\mathcal{O}_{\mathcal{X}}}(K, P)$$

Let x be an object of $\mathcal{X}$ which lies over an affine scheme $U = p(x)$. By Cohomology on Sites, Lemma 08JA we have the first equality in

$$R\Gamma(x, H) = R\,\mathrm{Hom}_{\mathcal{O}_{\mathcal{X}}}(K|_{\mathcal{X}/x}, P|_{\mathcal{X}/x}) = R\,\mathrm{Hom}_{\mathcal{O}}(K|_{\mathcal{X}_{affine}/x}, P|_{\mathcal{X}_{affine}/x})$$

The second equality stems from the fact that the topos of the site $\mathcal{X}/x$ is equivalent to the topos of the site $\mathcal{X}_{affine}/x$, see Sheaves on Stacks, Equation (0H0C). We may write $K = \epsilon^* N$ for some N in $QC(\mathcal{O})$. Then by Cohomology on Sites, Lemma 0H0T we see that

$$R\Gamma(x, H) = R\,\mathrm{Hom}_{D(\mathcal{O}(x))}(R\Gamma(x, N), R\Gamma(x, P))$$

By Lemma 0H13 we see that $R\Gamma(x, P) = 0$ if $U \to \mathcal{X}$ is flat and hence $R\Gamma(x, H) = 0$ under the same hypothesis. By Lemma 0H14 we conclude that $R\Gamma(\mathcal{X}, H) = 0$ and therefore $\mathrm{Hom}(K, P) = 0$.

Step 3: $\mathcal{A}$ is contained in $Q_{\mathcal{X}}$. Let K be an object of $\mathcal{A}$ and let $x \to x'$ be a morphism of $\mathcal{X}_{affine}$. We have to show that

$$R\Gamma(x', K) \otimes^{\mathbf{L}}_{\mathcal{O}(x')} \mathcal{O}(x) \longrightarrow R\Gamma(x, K)$$

is a quasi-isomorphism, see footnote in statement of Cohomology on Sites, Lemma 0GZS. By the proof of Lemma 07BA and the discussion in Remark 0H11 we see that $\mathcal{A}$ is the image of the restriction of $Lg_!$ to $D_{QCoh}(\mathcal{O}_{\mathcal{X}_{flat,fppf}})$. Thus we may assume $K = Lg_! M$ for some M in $D_{QCoh}(\mathcal{O}_{\mathcal{X}_{flat,fppf}})$. Then the desired equality follow from Lemma 0H15. □